

ONLINE RETAIL STORE MANAGEMENT (SQL SERVER)

Purpose of this Document:

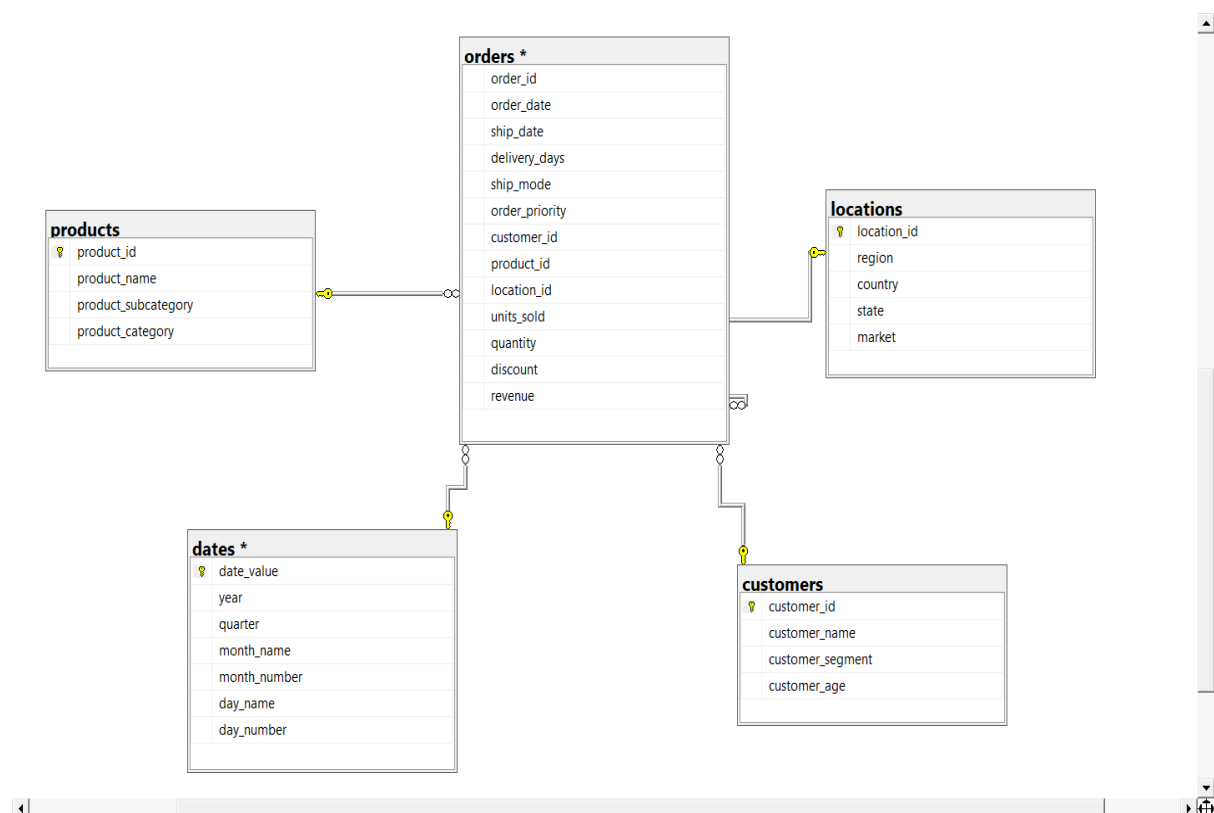
PDF is specifically created to showcase screenshots of my work, providing a step-by-step look at my SQL implementation. It includes:

- Database Design: Visual evidence of the 3rd Normal Form (3NF) architecture.
- Technical Implementation: Snapshots of advanced SQL features like Triggers, Indexes, and Stored Procedures.
- Analytical Results: Direct outputs from complex business queries, including revenue trends and customer rankings.

By reviewing these screenshots, you will see the practical application of my data and analytical skills within a realistic business context

Database Schema:

Entity-Relationship Diagram (ERD) showing the relationship between fact and dim tables.



CREATING DATABASE:

Creating a database named online retail store management and a raw data table containing named store data that contain cleaned unnormalized bulk data from excel files.

Query:

```

1]
2  --Creating a database for an online shopping management system
3]
4
5  create database online_retail_store_management;
6  use online_retail_store_management;
7
8]
9  -----
10] --Creating a raw dataset table representing unnormalized online shopping data
11]
12]
13] CREATE TABLE store_data
14] (
15]     order_id VARCHAR(50),
16]     order_date DATE,
17]     ship_date DATE,
18]     ship_mode VARCHAR(50),
19]     customer_name VARCHAR(100),
20]     segment VARCHAR(50),
21]     customer_age INT,
22]     state VARCHAR(100),
23]     country VARCHAR(100),
24]     market VARCHAR(100),
25]     region VARCHAR(100),
26]     category VARCHAR(100),
27]     sub_category VARCHAR(100),
28]     product_name VARCHAR(255),
29]     units_sold varchar(20),
30]     quantity INT,
31]     discount varchar(20),
32]     order_priority VARCHAR(20)
33] );
34]
35] -----
36] --Inserting data in the created table

```

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Output:

order_id	order_date	ship_date	ship_mode	customer_name	segment	customer_age	state	country	market	region	category	sub_category
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BULK INSERTION OF DATA AND CHECKING FOR NULLS IF ANY:

Inserting bulk data 1 lakh + rows in the created store data table and checking for nulls if any.

Query:

```
34 |  
35 | --Inserting data in the created table  
36 |  
37 |  
38 | BULK INSERT store_data  
39 | FROM 'C:\Users\Aditya\Desktop\SuperStoreOrders - SuperStoreOrders 3.csv'  
40 | WITH (  
41 |     FIRSTROW = 2,  
42 |     FORMAT = 'CSV',  
43 |     FIELDTERMINATOR = ',',  
44 |     FIELDQUOTE = '''',  
45 |     ROWTERMINATOR = '\n',  
46 |     TABLOCK  
47 | );  
48 |  
49 |  
50 | -- Checking for nulls if any  
51 |  
52 |  
53 | select  
54 | sum(case when order_id is null then 1 else 0 end) as order_id_nulls,  
55 | sum(case when order_date is null then 1 else 0 end) as order_date_nulls,  
56 | sum(case when ship_date is null then 1 else 0 end) as ship_date_nulls,  
57 | sum(case when ship_mode is null then 1 else 0 end) as ship_mode_nulls,  
58 | sum(case when customer_name is null then 1 else 0 end) as customer_name_nulls,  
59 | sum(case when segment is null then 1 else 0 end) as segment_nulls,  
60 | sum(case when customer_age is null then 1 else 0 end) as customer_age_nulls,  
61 | sum(case when state is null then 1 else 0 end) as state_nulls,  
62 | sum(case when country is null then 1 else 0 end) as country_nulls,  
63 | sum(case when market is null then 1 else 0 end) as market_nulls,  
64 | sum(case when region is null then 1 else 0 end) as region_nulls,  
65 | sum(case when category is null then 1 else 0 end) as category_nulls,  
66 | sum(case when sub_category is null then 1 else 0 end) as sub_category_nulls,  
67 | sum(case when product_name is null then 1 else 0 end) as product_name_nulls,  
68 | sum(case when units_sold is null then 1 else 0 end) as units_sold_nulls,  
69 | sum(case when quantity is null then 1 else 0 end) as quantity_nulls,  
70 |
```

Output:

Output for bulk insertion

Results															Messages
	order_id	order_date	ship_date	ship_mode	customer_name	segment	customer_age	state	country	market	region	category	sub_category	product_name	
1	AG-2011-2040	2011-01-01	2011-01-06	Standard Class	Toby Braunhardt	Consumer	47	Constantine	Algeria	Africa	Africa	Office Supplies	Storage	Tenex Lockers, Blue	
2	IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	64	New South Wales	Australia	APAC	Oceania	Office Supplies	Supplies	Acme Trimmer, High Sp	
3	HU-2011-1220	2011-01-01	2011-01-05	Second Class	Annie Thurman	Consumer	45	Budapest	Hungary	EMEA	EMEA	Office Supplies	Storage	Tenex Box, Single Width	
4	IT-2011-3647632	2011-01-01	2011-01-05	Second Class	Eugene Moren	Home Office	32	Stockholm	Sweden	EU	North	Office Supplies	Paper	Enemmax Note Cards, Pn	
5	IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	66	New South Wales	Australia	APAC	Oceania	Furniture	Furnishings	Eldon Light Bulb, Duo Pt	
6	IN-2011-47883	2011-01-01	2011-01-08	Standard Class	Joseph Holt	Consumer	57	New South Wales	Australia	APAC	Oceania	Office Supplies	Paper	Eaton Computer Printout	
7	CA-2011-1510	2011-01-02	2011-01-06	Standard Class	Magdelene Morse	Consumer	43	Ontario	Canada	Canada	Canada	Technology	Machines	Okidata Inkjet, Wireless	
8	IN-2011-79397	2011-01-03	2011-01-03	Same Day	Kean Nguyen	Corporate	50	New South Wales	Australia	APAC	Oceania	Office Supplies	Appliances	Hoover Microwave, Whit	
9	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	Consumer	44	Auckland	New Zealand	APAC	Oceania	Technology	Copiers	Hewlett Wireless Fax, La	
10	IZ-2011-4680	2011-01-03	2011-01-07	Standard Class	Lindsay Williams	Corporate	59	Ninawa	Iraq	EMEA	EMEA	Furniture	Chairs	Novimex Swivel Stool, St	
11	IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks	Consumer	37	National Capital	Philippines	APAC	Southeast Asia	Office Supplies	Storage	Tenex Lockers, Industria	
12	IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks	Consumer	48	National Capital	Philippines	APAC	Southeast Asia	Furniture	Tables	Chromcraft Round Table	
13	ES-2011-4869686	2011-01-03	2011-01-07	Standard Class	Dorothy Dickinson	Consumer	70	England	United Kingdom	EU	North	Furniture	Bookcases	Dania Corner Shelving, T	
14	IN-2011-33652	2011-01-03	2011-01-09	Standard Class	Dennis Pardue	Home Office	32	Sarawak	Malaysia	APAC	Southeast Asia	Technology	Copiers	Hewlett Fax and Copier,	
15	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	Consumer	43	Auckland	New Zealand	APAC	Oceania	Furniture	Chairs	Hon Rocking Chair, Set c	
16	MX-2011-160234	2011-01-03	2011-01-07	Standard Class	Stewart Visinsky	Consumer	31	Guatemala	Guatemala	LATAM	Central	Technology	Phones	Nokia Headset, VoIP	
17	IR-2011-770	2011-01-03	2011-01-07	Standard Class	Jas O'Carroll	Consumer	48	Yazd	Iran	EMEA	EMEA	Office Supplies	Appliances	Breville Coffee Grinder, E	
18	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	Consumer	42	Auckland	New Zealand	APAC	Oceania	Technology	Accessories	Belkin Numeric Keypad, K	
19	ID-2011-80230	2011-01-03	2011-01-09	Standard Class	Ken Lonsdale	Consumer	70	Auckland	New Zealand	APAC	Oceania	Furniture	Chairs	SAFCO Chairmat, Black	
20	ID-2011-12596	2011-01-03	2011-01-08	Standard Class	Chris McAfee	Consumer	29	Nakhon Ratcha...	Thailand	APAC	Southeast Asia	Office Supplies	Storage	Smead File Cart, Blue	
21	IN-2011-79397	2011-01-03	2011-01-03	Same Day	Kean Nguyen	Corporate	66	New South Wales	Australia	APAC	Oceania	Office Supplies	Labels	Avery Color Coded Label	
22	IR-2011-7690	2011-01-03	2011-01-08	Second Class	Nat Gilpin	Corporate	59	Razavi Khorasan	Iran	EMEA	EMEA	Office Supplies	Art	BIC Sketch Pad, Water C	
23	IR-2011-770	2011-01-03	2011-01-07	Standard Class	Jas O'Carroll	Consumer	35	Yazd	Iran	EMEA	EMEA	Office Supplies	Storage	Rogers Folders, Industr	
24	TZ-2011-7370	2011-01-03	2011-01-08	Standard Class	Jack Garza	Consumer	56	Dar Es Salaam	Tanzania	Africa	Africa	Office Supplies	Supplies	Stiletto Scissors, Serrate	
25	IZ-2011-4680	2011-01-03	2011-01-07	Standard Class	Lindsay Williams	Corporate	57	Ninawa	Iraq	EMEA	EMEA	Office Supplies	Envelopes	Cameo Interoffice Envelc	
26	IN-2011-65159	2011-01-03	2011-01-07	Second Class	Larry Blacks	Consumer	59	National Capital	Philippines	APAC	Southeast Asia	Office Supplies	Fasteners	Stockwell Staples, Metal	
27	IR-2011-770	2011-01-03	2011-01-07	Standard Class	Jas O'Carroll	Consumer	49	Yazd	Iran	EMEA	EMEA	Office Supplies	Fasteners	Advantus Rubber Bands,	
28	MX-2011-111255	2011-01-04	2011-01-09	Second Class	Russell Applegate	Consumer	27	Parana	Brazil	LATAM	South	Furniture	Bookcases	Dania Classic Bookcase,	
29	MX-2011-140641	2011-01-04	2011-01-09	Standard Class	Maya Herman	Corporate	57	Yucat'in	Mexico	LATAM	North	Technology	Accessories	Enemmax Keyboard, Blue	
30	MX-2011-158771	2011-01-04	2011-01-11	Standard Class	Beth Thompson	Home Office	50	Granma	Cuba	LATAM	Caribbean	Office Supplies	Envelopes	Jiffy Interoffice Envelope,	
31	ES-2011-4939443	2011-01-04	2011-01-08	Standard Class	Arthur Pritchep	Consumer	66	Rh'ne-Alpes	France	EU	Central	Office Supplies	Art	Binney & Smith Sketch P	
32	MX-2011-111255	2011-01-04	2011-01-09	Second Class	Russell Applegate	Consumer	34	Parana	Brazil	LATAM	South	Office Supplies	Supplies	Fiskars Letter Opener, Ei	
33	MX-2011-140641	2011-01-04	2011-01-09	Standard Class	Maya Herman	Corporate	34	Yucat'in	Mexico	LATAM	North	Technology	Copiers	Sharp Ink, Laser	
34	US-2011-136007	2011-01-04	2011-01-11	Standard Class	Beth Thompson	Home Office	42	Alagoas	Brazil	LATAM	South	Office Supplies	Envelopes	Jiffy Interoffice Envelope,	
35	MX-2011-159373	2011-01-04	2011-01-08	Standard Class	Arthur Wiediger	Home Office	36	Santiago de Cuba	Cuba	LATAM	Caribbean	Furniture	Chairs	SAFCO Chairmat, Black	

Output for null check

	order_id_nulls	order_date_nulls	ship_date_nulls	ship_mode_nulls	customer_name_nulls	segment_nulls	customer_age_nulls	state_nulls	country_nulls	market_nulls	region_nulls	category
1	0	0	0	0	0	0	0	0	0	0	0	0

NORMALISATION:

By Normalizing the raw data into specialized tables like Customers, Products, and locations, I ensured that each piece of information is stored in only one place. Using Primary Keys and Foreign Keys I also established a clear relationship between entities.

```
76 -----
77 --Normalizing the bulk data into 3NF form
78 -----
79
80 -- creating customer table (dim)
81
82 create table customers
83 (
84     customer_id int identity(10001,1) primary key,
85     customer_name varchar(50),
86     customer_segment varchar(20),
87     customer_age int
88 );
89
90 -- creating products table (dim)
91
92 create table products
93 (
94     product_id int identity(50001,1) primary key,
95     product_name varchar(200),
96     product_subcategory varchar(30),
97     product_category varchar(30)
98 );
99
100 -- creating location table (dim)
101
102 create table locations
103 (
104     location_id int identity(1234,1) primary key,
105     region varchar(80),
106     country varchar(80),
107     state varchar(80),
108     market varchar(80)
109 );
110
```

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```

109 );
110
111 -- creating dates table (dim)
112
113 create table dates
114 (
115     date_value date primary key,
116     year int,
117     quarter varchar(2),
118     month_name varchar(20),
119     month_number int,
120     day_name varchar(20),
121     day_number int
122 );
123
124 -- creating orders table (fact)
125
126 create table orders
127 (
128     order_id varchar(20),
129     order_date date,
130     ship_date date,
131     delivery_days int,
132     ship_mode varchar(20),
133     order_priority varchar(20),
134     customer_id int,
135     foreign key (customer_id)
136     references customers(customer_id),
137     product_id int,
138     foreign key (product_id)
139     references products(product_id),
140     location_id int,
141     foreign key (location_id)
142     references locations(location_id),
143     units_sold int
144 );

```

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INSERTING DATA IN NORMALIZED TABLES FROM STORE DATA

After creating the above tables and establishing the relationship through primary and foreign keys I achieved the desired relationship structure but now it's time to insert data from store data table to the normalized structured I created.

Query:

```

149
150 --Inserting data from the unnormalized store_data table to normalized tables
151
152
153 --Inserting data in products table
154
155 insert into products (product_name , product_subcategory , product_category)
156 select distinct product_name , sub_category , category from store_data;
157
158 --Inserting data in customers table
159
160 insert into customers (customer_name , customer_segment , customer_age)
161 select distinct customer_name , segment , customer_age from store_data;
162
163 --Inserting data in locations table
164
165 insert into locations (region , country , state , market)
166 select distinct region , country , state , market from store_data;
167
168 --Inserting data in dates table
169
170 DECLARE @StartDate DATE = '2011-01-01';
171 DECLARE @EndDate DATE = '2014-12-31';
172 WHILE @StartDate <= @EndDate
173 BEGIN
174     INSERT INTO dates(date_value)
175     VALUES (@StartDate);
176
177     SET @StartDate = DATEADD(DAY, 1, @StartDate);
178 END;
179
180 UPDATE dates
181 SET
182     year = YEAR(date_value),
183     month_number = MONTH(date_value)

```

100% No issues found Ln: 73, Ch: 1 SPC CRLF UTF-8

```
176
177     SET @StartDate = DATEADD(DAY, 1, @StartDate);
178 END;
179
180 UPDATE dates
181 SET
182     year = YEAR(date_value),
183     month_number = MONTH(date_value),
184     month_name = DATENAME(MONTH, date_value),
185     quarter = concat('Q', DATEPART(QUARTER, date_value)),
186     day_number = DAY(date_value),
187     day_name = DATENAME(WEEKDAY, date_value);
188
189 --Inserting data in orders table
190
191 INSERT INTO orders
192 (order_id, order_date, ship_date, delivery_days, ship_mode, order_priority, customer_id, product_id, location_id, units_sold, quar
193 SELECT sd.order_id, sd.order_date, sd.ship_date, DATEDIFF(day, sd.order_date, sd.ship_date), sd.ship_mode,
194 sd.order_priority, c.customer_id, p.product_id, l.location_id, sd.units_sold, sd.quantity, sd.discount,
195 sd.units_sold * sd.quantity * (1 - CAST(REPLACE(sd.discount, '%', '' ) AS DECIMAL(5,2)) / 100)
196 FROM store_data AS sd
197 JOIN customers AS c
198 ON c.customer_name = sd.customer_name
199 AND c.customer_segment = sd.segment
200 AND c.customer_age = sd.customer_age
201 JOIN products AS p
202 ON p.product_name = sd.product_name
203 AND p.product_category = sd.category
204 AND p.product_subcategory = sd.sub_category
205 JOIN locations AS l
206 ON l.region = sd.region
207 AND l.country = sd.country
208 AND l.state = sd.state
209 AND l.market = sd.market;
210
```

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Output:

Inserted data in orders table

	order_id	order_date	ship_date	delivery_days	ship_mode	order_priority	customer_id	product_id	location_id	units_sold	quantity	discount	revenue
1	AU-2013-9920	2013-10-22	2013-10-29	7	Standard Class	Medium	24503	53080	1690	74	6	0.00%	444
2	ES-2014-37...	2014-02-07	2014-02-11	4	Standard Class	Medium	18874	53080	2267	87	7	0.00%	609
3	EG-2014-7000	2014-06-18	2014-06-20	2	Second Class	High	22402	53080	1305	12	1	0.00%	12
4	MX-2014-16...	2014-06-06	2014-06-13	7	Standard Class	Low	17705	53080	2006	33	4	0.00%	132
5	ES-2011-37...	2011-03-14	2011-03-16	2	First Class	Critical	19699	53081	1566	43	3	0.00%	129
6	MD-2011-82...	2011-02-10	2011-02-14	4	Standard Class	Medium	17626	53081	1793	14	1	0.00%	14
7	EG-2011-1800	2011-04-14	2011-04-19	5	Standard Class	High	21495	53081	1302	86	6	0.00%	516
8	NI-2011-7970	2011-10-26	2011-11-01	6	Standard Class	Medium	23444	53081	1388	4	1	70.00%	1
9	ES-2012-19...	2012-01-18	2012-01-20	2	Second Class	High	19116	53081	1555	43	3	0.00%	129
10	HU-2012-8780	2012-04-03	2012-04-08	5	Standard Class	Medium	17612	53081	1727	29	2	0.00%	58
11	CG-2014-58...	2014-06-03	2014-06-10	7	Standard Class	Low	23179	53081	1293	14	1	0.00%	14
12	TZ-2014-5290	2014-06-18	2014-06-22	4	Standard Class	Medium	21450	53081	1438	14	1	0.00%	14
13	IN-2012-40897	2012-04-06	2012-04-11	5	Second Class	Medium	22166	53084	1651	227	5	0.00%	1135
14	IT-2012-553...	2012-08-16	2012-08-22	6	Standard Class	Medium	17853	53084	1559	41	1	10.00%	36
15	IN-2013-49318	2013-06-03	2013-06-08	5	Second Class	Medium	22747	53084	2029	91	2	0.00%	182
16	IT-2013-333...	2013-07-24	2013-07-28	4	Standard Class	High	20933	53084	1610	318	7	0.00%	2226
17	ES-2014-20...	2014-03-04	2014-03-09	5	Standard Class	Medium	23571	53084	2022	273	6	0.00%	1638
18	ID-2014-52902	2014-12-30	2015-01-01	2	First Class	High	20149	53084	2344	113	3	17.00%	281
19	ES-2011-41...	2011-05-04	2011-05-08	4	Standard Class	High	17848	53085	1566	229	5	0.00%	1145
20	SF-2011-9140	2011-12-26	2011-12-27	1	First Class	High	22955	53085	1429	46	1	0.00%	46
21	ES-2012-44...	2012-05-21	2012-05-25	4	Second Class	High	21871	53085	2257	137	3	0.00%	411
22	MX-2012-15...	2012-12-21	2012-12-22	1	First Class	High	18861	53085	1575	122	4	0.00%	488

Inserted data in products table

	product_id	product_name	product_subcategory	product_category
1	50001	"While you Were Out" Message Book, One Form per ...	Paper	Office Supplies
2	50002	#10 Gummed Flap White Envelopes, 100/Box	Envelopes	Office Supplies
3	50003	#10 Self-Seal White Envelopes	Envelopes	Office Supplies
4	50004	#10 White Business Envelopes, 4 1/8 x 9 1/2	Envelopes	Office Supplies
5	50005	#10- 4 1/8" x 9 1/2" Recycled Envelopes	Envelopes	Office Supplies
6	50006	#10- 4 1/8" x 9 1/2" Security-Tint Envelopes	Envelopes	Office Supplies
7	50007	#10-4 1/8" x 9 1/2" Premium Diagonal Seam Envelopes	Envelopes	Office Supplies
8	50008	#6 3/4 Gummed Flap White Envelopes	Envelopes	Office Supplies
9	50009	1.7 Cubic Foot Compact "Cube" Office Refrigerators	Appliances	Office Supplies
10	50010	1/4 Fold Party Design Invitations & White Envelopes, 2...	Paper	Office Supplies
11	50011	12 Colored Short Pencils	Art	Office Supplies
12	50012	12-1/2 Diameter Round Wall Clock	Furnishings	Furniture
13	50013	14-7/8 x 11 Blue Bar Computer Printout Paper	Paper	Office Supplies
14	50014	2300 Heavy-Duty Transfer File Systems by Perma	Storage	Office Supplies
15	50015	24 Capacity Maxi Data Binder Racks, Pearl	Storage	Office Supplies
16	50016	24-Hour Round Wall Clock	Furnishings	Furniture
17	50017	3.6 Cubic Foot Counter Height Office Refrigerator	Appliances	Office Supplies
18	50018	36X48 HARDFLOOR CHAIRMAT	Furnishings	Furniture
19	50019	3D Systems Cube Printer, 2nd Generation, Magenta	Machines	Technology
20	50020	3D Systems Cube Printer, 2nd Generation, White	Machines	Technology
21	50021	3M Hanners With Command Adhesive	Furnishings	Furniture

Inserted data in customer table

	customer_id	customer_name	customer_segment	customer_age
1	10001	Mitch Willingham	Corporate	36
2	10002	Candace McMahon	Corporate	42
3	10003	Duane Noonan	Consumer	61
4	10004	Mike Caudle	Corporate	43
5	10005	Muhammed Yedwab	Corporate	46
6	10006	Kean Thornton	Consumer	55
7	10007	Karen Bern	Corporate	34
8	10008	Pamela Stobb	Consumer	29
9	10009	Alan Barnes	Consumer	30
10	10010	Jim Epp	Corporate	57
11	10011	Charles Sheldon	Corporate	34
12	10012	Bryan Mills	Consumer	32
13	10013	John Grady	Corporate	24
14	10014	Victoria Pisteka	Corporate	67
15	10015	Luke Schmidt	Corporate	36
16	10016	Deirdre Greer	Corporate	35
17	10017	Manibeth Dona	Consumer	59
18	10018	Art Ferguson	Consumer	38
19	10019	Trudy Brown	Consumer	32
20	10020	Matt Connell	Corporate	70
21	10021	Greg Guthrie	Corporate	38
22	10022	Sally Knutson	Consumer	55

Inserted data in location table

	location_id	region	country	state	market
1	1234	Africa	Algeria	Alger	Africa
2	1235	Africa	Algeria	Annaba	Africa
3	1236	Africa	Algeria	Batna	Africa
4	1237	Africa	Algeria	Bechar	Africa
5	1238	Africa	Algeria	Bejaia	Africa
6	1239	Africa	Algeria	Constantine	Africa
7	1240	Africa	Algeria	Djelfa	Africa
8	1241	Africa	Algeria	Guelma	Africa
9	1242	Africa	Algeria	Laghouat	Africa
10	1243	Africa	Algeria	Medea	Africa
11	1244	Africa	Algeria	Mostaganem	Africa
12	1245	Africa	Algeria	Oran	Africa
13	1246	Africa	Algeria	Saida	Africa
14	1247	Africa	Algeria	Sidi Bel Abbes	Africa
15	1248	Africa	Algeria	Skikda	Africa
16	1249	Africa	Algeria	Tipaza	Africa
17	1250	Africa	Algeria	Tizi Ouzou	Africa
18	1251	Africa	Algeria	Tlemcen	Africa
19	1252	Africa	Angola	Benguela	Africa
20	1253	Africa	Angola	Bie	Africa
21	1254	Africa	Angola	Huambo	Africa

Inserted data in date table

	date_value	year	quarter	month_name	month_number	day_name	day_number
1	2011-01-01	2011	Q1	January	1	Saturday	1
2	2011-01-02	2011	Q1	January	1	Sunday	2
3	2011-01-03	2011	Q1	January	1	Monday	3
4	2011-01-04	2011	Q1	January	1	Tuesday	4
5	2011-01-05	2011	Q1	January	1	Wednesday	5
6	2011-01-06	2011	Q1	January	1	Thursday	6
7	2011-01-07	2011	Q1	January	1	Friday	7
8	2011-01-08	2011	Q1	January	1	Saturday	8
9	2011-01-09	2011	Q1	January	1	Sunday	9
10	2011-01-10	2011	Q1	January	1	Monday	10
11	2011-01-11	2011	Q1	January	1	Tuesday	11
12	2011-01-12	2011	Q1	January	1	Wednesday	12
13	2011-01-13	2011	Q1	January	1	Thursday	13
14	2011-01-14	2011	Q1	January	1	Friday	14
15	2011-01-15	2011	Q1	January	1	Saturday	15
16	2011-01-16	2011	Q1	January	1	Sunday	16
17	2011-01-17	2011	Q1	January	1	Monday	17
18	2011-01-18	2011	Q1	January	1	Tuesday	18
19	2011-01-19	2011	Q1	January	1	Wednesday	19
20	2011-01-20	2011	Q1	January	1	Thursday	20
21	2011-01-21	2011	Q1	January	1	Friday	21
22	2011-01-22	2011	Q1	January	1	Saturday	22
23	2011-01-23	2011	Q1	January	1	Sunday	23

CREATING INDEXES FOR PERFORMANCE:

Created indexes on high-traffic columns such as region in the location table and category in the Products table to accelerate filtering

Query:

```
--Creating indexes on necessary columns to improve query performance
```

```
create index idx_order_product_id on orders(product_id);
create index idx_order_customer_id on orders(customer_id);
create index idx_order_location_id on orders(location_id);
create index idx_order_order_date on orders(product_id);
create index idx_product_category on products(product_category);
create index idx_country on locations(region);
```

```
select t.name AS table_name,
       i.name AS index_name,
       i.type_desc AS index_type,
       i.is_primary_key,
       i.is_unique
from sys.indexes i
join sys.tables t
  on i.object_id = t.object_id
where i.index_id > 0
order by t.name, i.name;
```

Output:

	table_name	index_name	index_type	is_primary_key	is_unique
1	customers	PK_customer_CD65CB85ECC67778	CLUSTERED	1	1
2	dates	PK_date_tab_92DCEC694591D0CE	CLUSTERED	1	1
3	locations	idx_country	NONCLUSTERED	0	0
4	locations	PK_location_771831EAC8BEE1C2	CLUSTERED	1	1
5	order_audit	PK_order_au_5AF33E33C058D6F3	CLUSTERED	1	1
6	products	idx_product_category	NONCLUSTERED	0	0
7	products	PK_products_47027DF584ABD87B	CLUSTERED	1	1
8	sysdiagram	PK_sysdiagram_C2B05B6176EF4720	CLUSTERED	1	1
9	sysdiagram	UK_principal_name	NONCLUSTERED	0	1

CREATING ANALYTICAL VIEWS:

Created a consolidated view that joins Orders, Customers, and products to provide a high-level snapshot of product analysis, customer demographics and shipping insights.

Query:

Product Performance View

```
-- Creating Analytical Views
```

```
-- Product Performance View
```

```
create view product_performance_view as
with product_summery as
(
select
  dense_rank() over (order by sum(o.revenue) desc) as product_rank,
  p.product_name, p.product_subcategory, p.product_category,
  sum(o.quantity) as total_quantity_sold,
  format(sum(o.revenue), 'N0') as total_revenue,
  count(distinct order_id) as Total_orders_placed
from products as p
join orders as o on
  p.product_id=o.product_id
group by product_name, product_subcategory, product_category
)
select * from product_summery
where product_rank <= 10;

select * from product_performance_view;
```


Customers demographics and segment insight’s view

```
-- Customer demographics and segment insights view

create view customer_segment_insights as
with CSI as
(
select
dense_rank() over (order by sum(o.revenue) desc) as Rank,
c.customer_segment as customer_segment,
CASE
WHEN c.customer_age < 25 THEN 'Under 25 (Gen Z)'
WHEN c.customer_age BETWEEN 25 AND 40 THEN '25-40 (Millennial)'
WHEN c.customer_age BETWEEN 41 AND 60 THEN '41-60 (Gen X)'
ELSE '60+ (Senior)'
END AS age_group,
count(distinct c.customer_id) as total_customers,
format(sum(o.revenue), 'N0') as total_revenue,
avg(o.revenue) as average_order_value
from customers as c
join orders as o on
c.customer_id=o.customer_id
group by customer_segment,
CASE
WHEN c.customer_age < 25 THEN 'Under 25 (Gen Z)'
WHEN c.customer_age BETWEEN 25 AND 40 THEN '25-40 (Millennial)'
WHEN c.customer_age BETWEEN 41 AND 60 THEN '41-60 (Gen X)'
ELSE '60+ (Senior)'
END
END
```

Shipping & logistics efficiency view

```
--shipping & logistics efficiency view

create view shipping_logistics as
with SL as
(
select
dense_rank() over (order by count(distinct order_id) desc) as rank,
ship_mode,
order_priority,
avg(delivery_days) as average_delivery_days,
max(delivery_days) as max_delivery_days,
format(count(distinct order_id), 'N0') as total_shipments
from orders
group by ship_mode, order_priority
)
select * from SL
where rank<=10;

select * from shipping_logistics;
```

Output:

Output for product Performance View

	product_rank	product_name	product_subcategory	product_category	total_quantity_sold	total_revenue	Total_orders_placed
1	1	Apple Smart Phone, Full Size	Phones	Technology	171	395,200	51
2	2	Cisco Smart Phone, Full Size	Phones	Technology	139	369,476	38
3	3	Motorola Smart Phone, Full Size	Phones	Technology	134	360,506	38
4	4	Nokia Smart Phone, Full Size	Phones	Technology	147	273,350	47
5	5	Hon Executive Leather Armcha...	Chairs	Furniture	169	263,382	49
6	6	Samsung Smart Phone, VoIP	Phones	Technology	97	251,948	23
7	7	SAFCO Executive Leather Arm...	Chairs	Furniture	111	233,411	28
8	8	Canon imageCLASS 2200 Adv...	Copiers	Technology	20	231,000	5
9	9	Hoover Stove, Red	Appliances	Office Supplies	62	226,407	15
10	10	Sauder Classic Bookcase, Tra...	Bookcases	Furniture	113	225,792	29

Output for customers demographics and segment insight's view

Results		Messages				
Rank	customer_segment	age_group	total_customers	total_revenue	average_order_value	
1	Consumer	41-60 (Gen X)	5851	10,713,916	1037	
2	Consumer	25-40 (Mille...	4727	9,102,572	1076	
3	Corporate	41-60 (Gen X)	3375	6,591,838	1089	
4	Consumer	60+ (Senior)	2893	5,711,994	1101	
5	Corporate	25-40 (Mille...	2690	5,166,041	1067	
6	Home Office	41-60 (Gen X)	2101	3,901,706	1055	
7	Corporate	60+ (Senior)	1697	3,428,382	1124	
8	Home Office	25-40 (Mille...	1670	3,312,489	1134	
9	Consumer	Under 25 (G...	1424	2,784,354	1089	
10	Home Office	60+ (Senior)	1022	1,755,439	976	

Output for shipping & logistics efficiency view

Results		Messages				
rank	ship_mode	order_priority	average_delivery_days	max_delivery_days	total_shipments	
1	Standard Class	Medium	5	7	10,648	
2	Standard Class	High	4	5	3,414	
3	Second Class	Medium	3	5	2,417	
4	Second Class	High	2	5	1,998	
5	First Class	High	2	3	1,754	
6	Standard Class	Low	6	7	1,212	
7	First Class	Medium	2	3	1,188	
8	First Class	Critical	2	3	896	
9	Second Class	Critical	2	3	720	
10	Same Day	High	0	1	642	

BUSINESS ANALYSIS QUERIES:

To demonstrate the practical utility of the database, I implemented analytical queries focusing on:

- **Revenue Performance:** Tracking monthly and yearly revenue distribution and growth % trends to identify opportunities.
- **Customer Lifetime Value:** Ranking customers by total spend using Window Functions to identify their lifetime spend with the company and other trends.
- **Product Analytics:** Identifying the top 5 and bottom 5 products to assist with inventory and marketing strategy.

Query:

Query for MOM% Analysis

```
-- Business analysis queries

-- MOM analysis

with monthly_analysis as
(
  select d.year,
         d.month_number,
         d.month_name,
         sum(o.revenue) as Total_revenue
  from dates as d
       join orders as o on
         d.date_value = o.order_date
  group by
    d.year,
    d.month_number,
    d.month_name
)

select year, month_name, total_revenue,
       isnull(lag(total_revenue) over (order by year, month_number asc), 0) as previous_month_revenue,
       isnull(cast(
         ((total_revenue - lag(total_revenue) over (order by year, month_number)) * 100.0
        ) /
        nullif(
          lag(total_revenue) over (order by year, month_number), 0
        ) as decimal(10,2)), 0) as [mom%]
  from monthly_analysis
 order by year, month_number;
```

Query for YOY% analysis

```
-- YOY analysis

with yearly_sales as
(
  select d.year,
         sum(o.revenue) as total_revenue
  from orders as o
       join dates as d
         on o.order_date = d.date_value
  group by d.year
)

select year,
       format(total_revenue,'n0') as total_revenue,
       isnull(format(lag(total_revenue) over(order by year),'n0'),0) as previous_year_revenue,
       concat(isnull(cast(
         ((total_revenue - lag(total_revenue) over(order by year)) * 100.0)
         /
         nullif(lag(total_revenue) over(order by year),0)as decimal(10,2) ),0),'%') as yoy_growth
from yearly_sales;
```

Query for top 5 products

```
-- Top 5 products

select top 5
       dense_rank() over(order by sum(o.revenue) desc) as product_rank,
       p.product_name,
       p.product_category,
       p.product_subcategory,
       sum(o.quantity) as total_quantity_sold,
       count(distinct o.order_id) as total_orders,
       format(sum(o.revenue),'n0') as total_revenue,
       format(avg(o.revenue),'n0') as average_order_value
from products as p
join orders as o
  on p.product_id = o.product_id
group by
  p.product_name,
  p.product_category,
  p.product_subcategory
order by sum(o.revenue) desc;
```

Query for bottom 5 products

```
-- Bottom 5 products

select top 5
       dense_rank() over(order by sum(o.revenue) desc) as product_rank,
       p.product_name,
       p.product_category,
       p.product_subcategory,
       sum(o.quantity) as total_quantity_sold,
       count(distinct o.order_id) as total_orders,
       format(sum(o.revenue),'n0') as total_revenue,
       format(avg(o.revenue),'n0') as average_order_value
from products as p
join orders as o
  on p.product_id = o.product_id
group by
  p.product_name,
  p.product_category,
  p.product_subcategory
order by sum(o.revenue) asc;
```

Query for customer life time value

```
-- Customer lifetime value

with customer_clv as
(
  select
    c.customer_id,
    c.customer_name,
    c.customer_segment,
    count(distinct o.order_id) as total_orders,
    sum(o.revenue) as lifetime_value,
    avg(o.revenue) as average_order_value
  from customers as c
  join orders as o
    on c.customer_id = o.customer_id
  group by
    c.customer_id,
    c.customer_name,
    c.customer_segment
)

select
  dense_rank() over(order by lifetime_value desc) as customer_rank,
  customer_name,
  customer_segment,
  total_orders,
  format(lifetime_value,'n0') as lifetime_value,
  format(average_order_value,'n0') as average_order_value
from customer_clv
order by lifetime_value desc;
```

Output:

Output for MOM% Analysis

	year	month_name	total_revenue	previous_month_revenue	mom%
1	2011	January	418271	0	0.00
2	2011	February	448783	418271	7.29
3	2011	March	596138	448783	32.83
4	2011	April	506371	596138	-15.06
5	2011	May	662307	506371	30.79
6	2011	June	883836	662307	33.45
7	2011	July	460029	883836	-47.95
8	2011	August	952389	460029	107.03
9	2011	September	1295177	952389	35.99
10	2011	October	854556	1295177	-34.02
11	2011	November	1329443	854556	55.57
12	2011	December	1446496	1329443	8.80
13	2012	January	591561	1446496	-59.10
14	2012	February	448292	591561	-24.22
15	2012	March	733074	448292	63.53
16	2012	April	694891	733074	-5.21
17	2012	May	902011	694891	29.81
18	2012	June	974502	902011	8.04
19	2012	July	629750	974502	-35.38
20	2012	August	1437246	629750	128.22
21	2012	September	1241945	1437246	-13.59
22	2012	October	1090995	1241945	-12.15
23	2012	November	1346623	1090995	23.43
24	2012	December	1495014	1346623	11.02
25	2013	January	938718	1495014	-37.21
26	2013	February	811148	938718	-13.59
27	2013	March	865033	811148	6.64
28	2013	April	711966	865033	-17.69
29	2013	May	1127210	711966	58.32
30	2013	June	1816502	1127210	61.15

Output for YOY% analysis

	year	total_revenue	previous_year_revenue	yoy_growth
1	2011	9,853,796	0	0.00%
2	2012	11,585,904	9,853,796	17.58%
3	2013	14,990,987	11,585,904	29.39%
4	2014	18,768,789	14,990,987	25.20%

Output for top 5 products

	product_rank	product_name	product_category	product_subcategory	total_quantity_sold	total_orders	total_revenue	average_order_value
1	1	Apple Smart Phone, Full Size	Technology	Phones	171	51	395,200	7,749
2	2	Cisco Smart Phone, Full Si...	Technology	Phones	139	38	369,476	9,723
3	3	Motorola Smart Phone, Ful...	Technology	Phones	134	38	360,506	9,487
4	4	Nokia Smart Phone, Full Si...	Technology	Phones	147	47	273,350	5,815
5	5	Hon Executive Leather Arm...	Furniture	Chairs	169	49	263,382	5,375

Output for bottom 5 products

	product_rank	product_name	product_category	product_subcategory	total_quantity_sold	total_orders	total_revenue	average_order_value
1	3301	Eureka Disposable Bags for Sanitaire Vibra Groom...	Office Supplies	Appliances	2	1	0	0
2	3300	Acco Economy Flexible Poly Round Ring Binder	Office Supplies	Binders	6	4	5	1
3	3299	Xerox 20	Office Supplies	Paper	1	1	6	6
4	3298	Chromcraft Training Table, Adjustable Height	Furniture	Tables	1	1	7	7
5	3297	Grip Seal Envelopes	Office Supplies	Envelopes	2	1	11	11

Output for customer life time value

	customer_rank	customer_name	customer_segment	total_orders	lifetime_value	average_order_value
1	1	Adrian Barton	Consumer	2	128,721	64,360
2	2	Patrick Jones	Corporate	1	111,426	111,426
3	3	Tamara Chand	Corporate	1	87,500	87,500
4	4	Elpida Rittenbach	Corporate	2	78,246	39,123
5	5	Carol Adams	Corporate	1	76,989	76,989
6	6	Harry Greene	Consumer	3	74,414	24,804
7	7	Dave Poirier	Corporate	1	74,214	74,214
8	8	Vivek Grady	Corporate	2	73,892	36,946
9	9	Sean Christens...	Consumer	1	72,576	72,576
10	10	Maria Bertelson	Consumer	1	69,136	69,136
11	11	Sean Miller	Home Office	2	68,053	22,684
12	12	Pauline Johnson	Consumer	2	66,610	33,305
13	13	Ann Steele	Home Office	2	64,404	32,202
14	14	Bill Shonely	Corporate	1	63,720	31,860
15	15	Patrick O'Brill	Consumer	3	62,101	20,700
16	16	David Kendrick	Corporate	3	59,845	19,948

CREATING STORE PROCEDURE:

Stored procedures are reusable SQL programs that automate queries, improve performance, and allow parameterized data retrieval.

- **Customer Purchase History:** Retrieves the purchase history of a specific customer.
- **Sales Between Two Dates:** Displays sales records within a user-defined date range.
- **Customer Churn Analysis:** Identifies inactive customers based on their last purchase and days since their last order.

Query:

Query for customer purchase history store procedure

```
-----  
-- Creating store procedures  
-----  
  
-- customer purchase history store procedure  
  
create procedure sp_customer_history @customer_id int as  
begin  
select o.order_id,  
       o.order_date,  
       p.product_name,  
       o.quantity,  
       o.revenue  
from orders o  
       join products p  
       on o.product_id = p.product_id  
where o.customer_id = @customer_id  
order by o.order_date;  
end;  
  
exec sp_customer_history 10001;
```

Query for sales between two dates store procedure

```
] -- Sales between two dates store procedure  
  
create procedure sp_sales_between_dates  
@start_date date,  
@end_date date as  
begin  
select count(distinct order_id) as total_orders,  
       sum(revenue) as total_revenue,  
       sum(quantity) as total_quantity  
from orders  
where order_date between @start_date and @end_date;  
  
end;  
  
exec sp_sales_between_dates '2012-01-01', '2012-12-31';
```

Query for customer churn analysis store procedure

```
-- Customer churn analysis store procedure  
  
create procedure sp_customer_churn_analysis  
@days int as  
begin  
select c.customer_id,  
       c.customer_name,  
       c.customer_segment,  
       max(o.order_date) as last_order_date,  
       datediff(day, max(o.order_date), '2014-12-31') as days_since_last_order,  
       sum(o.revenue) as lifetime_revenue  
from customers as c  
       join orders as o  
       on c.customer_id = o.customer_id  
group by c.customer_id,  
         c.customer_name,  
         c.customer_segment  
having datediff(day, max(o.order_date), getdate()) >= @days  
order by lifetime_revenue desc;  
end;  
  
exec sp_customer_churn_analysis 90;
```

Output:

Output for customer purchase history store procedure

(since I asked for a single customer id i.e. 10001 it is giving me records of that customer only)

	order_id	order_date	product_name	quantity	revenue
1	MX-2011-100790	2011-10-19	Boston Canvas, Fluorescent	5	915

Output for sales between two dates store procedure

(I asked for total sales between '2012-01-01' to '2012-12-31' so it's returning the same)

	total_orders	total_revenue	total_quantity
1	5343	11585904	38111

Output for customer churn analysis store procedure

(Since I asked for last 90 days customer churn for everyone its giving me the same)

#	Results	Messages	customer_id	customer_name	customer_segment	last_order_date	days_since_last_order	lifetime_revenue
1			12099	Adrian Barton	Consumer	2014-06-06	208	128721
2			22544	Patrick Jones	Corporate	2014-09-08	114	111426
3			18433	Tamara Chand	Corporate	2013-10-03	454	87500
4			26273	Elpida Rittenbach	Corporate	2011-09-08	1210	78246
5			23863	Carol Adams	Corporate	2011-02-10	1420	76989
6			19628	Harry Greene	Consumer	2014-11-19	42	74414
7			21313	Dave Poirier	Corporate	2014-10-28	64	74214
8			16371	Vivek Grady	Corporate	2014-09-26	96	73892
9			30617	Sean Christensen	Consumer	2012-08-02	881	72576
10			21410	Maria Bertelson	Consumer	2014-06-24	190	69136
11			22995	Sean Miller	Home Office	2014-10-16	76	68053
12			35273	Pauline Johnson	Consumer	2014-06-18	196	66610
13			30348	Ann Steele	Home Office	2013-12-16	380	64404
14			29541	Bill Shonely	Corporate	2013-04-17	623	63720
15			30485	Patrick O'Brill	Consumer	2014-08-24	129	62101
16			35377	David Kendrick	Corporate	2012-12-21	740	59845
17			34296	Raymond Buch	Consumer	2014-03-24	282	56000
18			11994	Denny Blanton	Consumer	2014-01-18	347	55539
19			33809	Michael Domin...	Corporate	2013-06-20	559	55370
20			36235	Patrick O'Donnell	Consumer	2014-01-14	351	54015
21			28527	Greg Tran	Consumer	2013-12-30	366	53861
22			25525	Eugene Hildebr...	Home Office	2014-10-23	69	53486
23			14793	Laurel Beltran	Home Office	2014-03-01	305	52894
24			31092	George Bell	Corporate	2014-09-13	109	52397
25			14688	Dean percer	Home Office	2013-09-25	462	52089
26			34191	Charles McCro...	Consumer	2014-11-12	49	51988
27			21165	Cynthia Arntzen	Consumer	2013-06-12	567	51768
28			19764	Mike Gockenba...	Consumer	2014-02-14	320	51572
29			34290	Marina Lichtens...	Corporate	2013-06-04	575	51534
30			13451	Daniel Raolin	Home Office	2014-11-13	48	50660

TRIGGER IMPLIMENTATION:

Automated Event Handling

- **Real-Time Auditing:** Developed specialized triggers for three distinct database events—Insert, Update, and Delete—to capture the lifecycle of an order.
- **Data Traceability:** Each trigger automatically records the action type, action date and time and order id.

- **Operational Integrity:** Demonstrated the use of triggers to maintain synchronization between transactional tables and audit logs, ensuring 100% accountability for data changes.

Query:

Creating order audit trigger table to store the insert update and delete actions performed on order table.

```
-- Creating a trigger table

create table order_audit
(
    audit_id int identity(1,1) primary key,
    order_id varchar(20),
    action_type varchar(10),
    action_date datetime default getdate()
);
```

Creating insert update and delete triggers

```
--
518 -- Insert trigger
519
520 create trigger insert_after_orders on orders
521 after insert as
522 begin
523     insert into order_audit
524     (order_id, action_type)
525     select order_id, 'insert'
526     from inserted;
527 end;
528
529 -- Update trigger
530
531 create trigger update_after_orders on orders
532 after update as
533 begin
534     insert into order_audit
535     (order_id, action_type)
536     select order_id, 'update'
537     from inserted;
538 end;
539
540 -- Delete Trigger
541
542 create trigger trg_orders_delete on orders
543 after delete as
544 begin
545     insert into order_audit
546     (order_id, action_type)
547     select order_id, 'delete'
548     from deleted;
549 end;
550
```

Testing the triggers created

```
551 -----
552 -- Testing created triggers
553 -----
554
555 -- testing insert trigger
556
557 insert into orders
558 (order_id, order_date, ship_date, delivery_days, ship_mode, order_priority, product_id, location_id, customer_id, units_sold, quantity, discount, revenue)
559 values ('TEST001', '2014-12-30', '2015-01-02', 3, 'Standard Class', 'Medium', 50001, 1234, 10001, 100, 2, '0%', 200);
560
561 -- testing update trigger
562
563 update orders
564 set revenue = 500
565 where order_id = 'TEST001';
566
567 -- testing delete trigger
568
569 delete
570 from orders
571 where order_id = 'TEST001';
572
573 select * from order_audit;
574
575
576
577
578
```

Output:

The output consists of the recorded actions performed on orders table stored in order audit trigger table to track actions performed.

	audit_id	order_id	action_type	action_date
1	1	TEST001	insert	2026-06-22 16:07:34.590
2	2	TEST001	update	2026-06-22 16:07:40.990
3	3	TEST001	delete	2026-06-22 16:07:47.063

-----X-----

